Problem 10.5

At a constant force of interest, \$200 accumulates to \$240 over the course of 8 years. Find the force of interest δ.

Solution

For a constant force of interest, the accumulation factor over t years is

$$e^{\delta t}.$$

We are given that \$200 accumulates to \$240 in 8 years, so

$$200e^{8\delta} = 240.$$

Divide by 200:

$$e^{8\delta} = 1.2.$$

Taking natural logarithms,

$$8\delta = \ln(1.2).$$

Thus,

$$\delta = \frac{1}{8}\ln(1.2).$$

Therefore,

$$\delta = 0.022790.$$

Hence, the force of interest is

$$\boxed{0.022790}.$$